

SEHAR ANWAR

Applicant – M.S. in Fluid Mechanics

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<https://www.linkedin.com/in/sehar-anwar-82b981427> | Final Year Research in Plasma Physics

OBJECTIVE

Physics graduate, from the University of Engineering and Technology (UET) with a strong foundation in plasma physics, classical mechanics, and computational methods, seeking to pursue a Master. Currently interning in an AI-focused role, building hands-on skills in machine learning tools and computational workflows applicable to numerical simulation and fluid dynamics research.

EDUCATION

University of Engineering and Technology (UET) Lahore, Pakistan [December 2022 – July 2026]

Bachelor of Science (B.S.) in Physics

CGPA: [3.3/4.0]

- Undergraduate Thesis: Comparison of two Cases of Double Beltrami Field in Electron-Ion Plasma

RELEVANT COURSEWORK

Classical Mechanics · Mechanics (Lab) · Heat and Thermodynamics · Thermal and Statistical Physics · Waves and Oscillations (+ Lab) · Electrodynamics I & II · Mathematical Methods of Physics I & II · Ordinary Differential Equations · Calculus I & II · Linear Algebra · Computational Physics · Statistical Methods in Physics · Physical and Geometrical Optics · Solid State Physics I & II · Electronics I & II (+ Lab) · Quantum Mechanics I & II · Nuclear Physics I & II

INTERNSHIP EXPERIENCE

SoftSolex

[August 2026 – Present]

AI Intern | Lahore, Pakistan

- Learning many tools like GitHub, VS code, Google Antigravity, Kaggle and Jupyter Notebook, Data processing and Pandas Workflows, displaying data with Libraries/Graphs using Matplotlib and Seaborn, etc.

Mechanical Internship

[June 2025 – August 2025]

HUBCO Power Plant, Narowal

- Gained hands-on exposure to fuel oil, cooling water, and condensate/feedwater piping systems, including pipe flow, pressure drop, and pump operation.
- Studied HRSG and steam turbine cycle operation, applying concepts of two-phase flow, heat exchanger performance, and thermodynamic cycles.
- Shadowed maintenance and operations teams on rotating equipment (pumps, engines, turbines), reinforcing fluid dynamics and thermodynamics coursework through real-world observation.

TECHNICAL SKILLS

Programming & Computation

- Python, MATLAB
- Machine learning tools/workflows (from AI internship)

Industrial Skills

- Technical report writing
- industrial fluid & piping systems
- HRSG/steam turbine operations

Mathematical Tools

- Differential Equations, Linear Algebra, Mathematical Methods of Physics

Laboratory Skills

- Mechanics Lab, Electronics Lab, Modern Physics Lab, Waves & Oscillations Lab

Languages

- English, Urdu, Punjabi

RESEARCH / PROJECT

- Final Year Research Thesis on Comparison of two Cases of Double Beltrami Field in Electron-Ion Plasma.
- Designed a rain-detection sensor system (Modern Physics Lab) that automatically retracts an outdoor clothesline under shelter when rain is detected and extends it back out once dry.

CERTIFICATE

- HUBCO Internship Certificate

SEMINAR/CONFERENCE

- **One Day International Symposium on "Mysteries of Quantum Mechanics"**

Gained deeper insight into foundational and unresolved questions in quantum mechanics and their broader implications for modern physics, including perspectives from international quantum optics research.

- **International Conference on Hydrology & Water Resources**

Gained deeper insight into foundational and unresolved questions in quantum mechanics and their broader implications for modern physics, including perspectives from international quantum optics research.

REFERENCES

Dr. Saba Majeed Gondal

Assistant Professor

Department of Physics

Main Campus, UET Lahore

<https://staff.uet.edu.pk/profile/571>

DR. Abdul Waheed Anwar

Associate Professor

Department of Physics

Main Campus, UET Lahore

<https://staff.uet.edu.pk/profile/443>